

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Bioscience and Biotechnology

2. Subject Name: Bioprocess Technology

L-T-P: 3-1-0

Credits: 4

3. Pre-requisites: Biochemical Reaction Engineering and Bioenergetics (BT20207)

4. Syllabus and reference books:

Syllabus:

1. Recombinant DNA technology processes and products using bacterial/yeast/animal cell cultures - Production and purification of recombinant microbial products of therapeutic and commercial interest - Recombinant human insulin, Tissue plasminogen activator (tPA), human serum albumin. Production and purification of antibiotics like Penicillin, Cephalosporins, Clavulanic acid.

2. Plant Biotechnology Processes and Products - Plant cell and tissue culture-based production of secondary metabolites of therapeutic and commercial interest.

3. Animal Biotechnology Processes and Products: Use of transgenic animals as bio-producers of therapeutic proteins, transgenic animal bioreactors, pharmaceutical potential of transgenic animals, quality control guidelines on transgenically generated bioactive materials; Vaccine production - Production and purification of live attenuated and inactivated vaccines, Classical and recombinant subunit vaccines, DNA vaccines, Process scale-up, Quality by design framework, Clinical trials.

4. Microbial and Enzyme Technology Processes and Products for Energy, Environment & Water:

(i) Global Energy crisis and Microbial Enhanced Oil Recovery (MEOR): EOR & MEOR Processes; MEOR Mechanisms; Ex-situ & In-situ MEOR; MEOR Agents - Biosurfactants and Biopolymers; MEOR modeling; Field trials - current status; Sustainability.

(ii) Bioethanol: First generation bioethanol from starch; Food versus fuel debate; Second generation bioethanol from lignocellulosic biomass; Biomass pretreatment; Enzymatic saccharification of cellulose and hemicellulose; Ethanol fermentation and recovery; Sustainability.

(iii) Biodiesel: First generation biodiesel from edible vegetable oils; Food versus fuel debate; Second generation biodiesel from non-conventional and non-edible plant oils; Third generation from algae feedstock; Transesterification - chemical and enzyme

catalyzed; Immobilized enzymes and Kinetics; Packed bed reactors; Biodiesel - Fuel characteristics; Sustainability.

(iv) Biorefinery: Petroleum refinery concept; Yeast, Lignocellulosic and Algal biomass feedstock based Biorefinery; Consolidated bioprocessing; Biofuels and Value-added products; Biooil and Biochar; Case studies; Sustainability.

Reference Books:

- 1) Process Biotechnology Fundamentals, by S. N. Mukhopadhyay
- 2) Development of Sustainable Bioprocesses: Modeling and Assessment by Elmar Heinzle, Arno P. Biwer and Charles L. Cooney (2006)
- 3) Bioprocess Engineering: Systems, Equipment and Facilities; by Bjorn K. Lydersen (Editor), Nancy A. D'Elia (Editor), Kim L. Nelson (Editor)
- 4) Biochemical Engineering and Biotechnology, by Ghasem D. Najafpour (Elsevier)
- 5) Animal Cell Culture and Technology by M. Butler

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Recombinant DNA technology processes and products using bacterial/yeast/animal cell cultures - Production and purification of recombinant microbial products of therapeutic and commercial interest - Recombinant human insulin, Tissue plasminogen activator (tPA), human serum albumin. Production and purification of antibiotics like Penicillin, Cephalosporins, Clavulanic acid	12
2.	Plant Biotechnology Processes and Products - Plant cell and tissue culture-based production of secondary metabolites of therapeutic and commercial interest	8
3.	Animal Biotechnology Processes and Products: Use of transgenic animals as bio-producers of therapeutic proteins, transgenic animal bioreactors, pharmaceutical potential of transgenic animals, quality control guidelines on transgenically generated bioactive materials; Vaccine production - Production and purification of live attenuated and inactivated vaccines, Classical and recombinant subunit vaccines, DNA vaccines, Process scale-up, Quality by design framework, Clinical trials.	12
4.	Global Energy crisis and Microbial Enhanced Oil Recovery (MEOR): EOR & MEOR Processes; MEOR	4

	Mechanisms; Ex-situ & In-situ MEOR; MEOR Agents - Biosurfactants and Biopolymers; MEOR modeling; Field trials - current status; Sustainability	
5.	Bioethanol: First generation bioethanol from starch; Food versus fuel debate; Second generation bioethanol from lignocellulosic biomass; Biomass pretreatment; Enzymatic saccharification of cellulose and hemicellulose; Ethanol fermentation and recovery; Sustainability	4
6.	Biodiesel: First generation biodiesel from edible vegetable oils; Food versus fuel debate; Second generation biodiesel from non-conventional and non-edible plant oils; Third generation from algae feedstock; Transesterification - chemical and enzyme catalyzed; Immobilized enzymes and Kinetics; Packed bed reactors; Biodiesel - Fuel characteristics; Sustainability	4
7.	Biorefinery: Petroleum refinery concept; Yeast, Lignocellulosic and Algal biomass feedstock based Biorefinery; Consolidated bioprocessing; Biofuels and Value added products; Biooil and Biochar; Case studies; Sustainability	4
Total number of hours		48